

CYNTHIA OWOLABI

Senior Software Engineer | Payments & Distributed Systems

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Portfolio & engineering case studies: adebolaowolabi32.github.io

Senior software engineer with 9+ years of experience across payments, distributed applications, cloud infrastructure and applied AI. Experience spans payment-provider integrations, accounting synchronization, compliance controls and cloud services at Duplo, Bitso, Microsoft, Indeed and Interswitch. Experience includes reducing API scan completion from 15 to 5 hours and delivering transit software for fare collection, ticketing and fleet tracking.

TECHNICAL FOCUS

Backend & data: TypeScript/Node.js, Java, Python, C#, Go, PHP/Laravel; PostgreSQL, MySQL, Redis, Cassandra; REST, Kafka

Frontend: React, Next.js, Angular, JavaScript, TypeScript

Cloud & delivery: Azure, AWS, Docker, Kubernetes, Terraform, CI/CD

Applied AI: LLM APIs, LangGraph, RAG, embeddings, vector databases

PROFESSIONAL EXPERIENCE

Duplo — Senior Full Stack Engineer

Jul 2025 – Present

Nigeria · Hybrid · Full-time

- Architect and develop internal administration systems using Next.js, TypeScript, NestJS and PostgreSQL, consolidating business and transaction data with boundary normalization and partial-availability handling.
- Implement Xero integrations for expenses, bills, invoices, payments, contacts and taxes, including synchronization jobs and endpoints; build partner registries and feature flags for partner-specific capabilities.
- Implement liquidity transfers, transaction-status and balance queries, and Wema V2 payouts, with partner-specific payment-provider and currency provisioning.
- Deliver Nigeria and South Africa compliance workflows with rejected-field resubmission and business activation controls; implement role-based access and maker-checker approvals for sensitive administrative actions.
- Build external-service mocks for accounting and payment integrations, enabling development and integration testing without continuous dependence on live third-party environments.

Bitso — Senior Software Engineer

Apr 2023 – Jul 2025

Mexico · Remote · Contract

- Developed and integrated an AI customer-support assistant using LLM APIs and LangGraph to answer natural-language queries.
- Designed agent workflows for transaction risk analysis, using semantic embeddings, vector retrieval and RAG to surface transaction information and flag suspicious patterns.
- Shipped Java and PHP/Laravel services for high-growth markets serving hundreds of thousands of users.

Microsoft — Senior Software Engineer

Dec 2021 – Feb 2023

Nigeria · Hybrid · Full-time

- Automated API scan processing, reducing completion time from 15 to 5 hours and saving 10 hours per scan.
- Automated runtime and documentation issue logging and background workflows using Azure DevOps, Service Bus, Durable Orchestration Functions, Worker Service and Application Insights.
- Built distributed web applications using C#, Python, TypeScript and Node.js, and implemented Azure CI/CD pipelines.

Indeed — Senior Software Engineer

Jun 2021 – Sep 2022

United States · Remote · Contract

- Maintained and upgraded an artifact-distribution system using Java, Go, Cassandra, MySQL, MongoDB, Consul and ZooKeeper.
- Migrated services from on-premise infrastructure to AWS using Python and upgraded services to run on Kubernetes, with Docker-based delivery.

Interswitch Group — Software Engineer

May 2018 – Nov 2021

Nigeria · Hybrid · Full-time

- Developed frontend and backend systems for urban mass transit: automated fare collection, ticketing and real-time fleet tracking.
- Delivered a web demonstration within six months, contributing to the company's first Lagos Metropolitan Area Transport Authority (LAMATA) contract.
- Refactored Enterprise Service Bus infrastructure into microservices, with reported uptime increasing from 88% to 99.7%.
- Developed an API gateway providing analytics, security and API management.

Zone — Software Engineer

Jan 2017 – Mar 2018

Nigeria · On-site · Full-time

- Architected and delivered software for the Deja-Vu Platform; performed code reviews, unit and integration testing, and produced system specifications and technical documentation.

Earlier: Embedded Systems Engineering Intern, ElectroHyve Research Institute of Technology · Jun–Nov 2015. Hardware development, component testing and system analysis.

SELECTED INDEPENDENT PROJECTS

LedgerDesk — Payment operations with independent approvals, PostgreSQL leased workers, provider reconciliation and durable attempt history.

RelayDesk — Artifact-delivery simulation and separate local HTTP worker lab with SHA-256 verification, durable SQLite jobs and recovery tests.

SourceDesk — Evidence-based support with pgvector retrieval, source snapshots and handoffs; 40 authored regression cases and five live smoke cases.

TrafficDesk — Deterministic transit simulation with disruption controls, passenger journeys and conservation tests.

EDUCATION & ACHIEVEMENTS

Federal University of Technology, Akure

BE, Electrical & Electronics Engineering · First Class Honours · CGPA 4.53/5

Gold Medal, American Mathematics Competition (2011) · Bronze Medal, International Junior Science Olympiad (2010) · Represented Nigeria, International Mathematical Olympiad (2011).